

HeROsim: graph-aware task placement on a heterogeneous serverless edge

What the simulator computes, how the labels are made, what the model is, how it is evaluated, and where the answer stands

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1 Summary

The question. A serverless platform on a cluster of heterogeneous edge devices receives inference requests from client devices and must place each on a replica (a compute platform on a machine). Does a graph neural network (GNN) that sees a batch of arriving tasks and the cluster *as a graph* place them better than (a) the industry-standard reactive scheduler (Knative: each task to the shortest reachable queue) and (b) a pointwise learned scorer that evaluates one (task, candidate) pair at a time? The pointwise model is a *control*: if it matches the GNN, the graph is not what the GNN is using.

The environment in three sentences. Five measured device platforms (Raspberry Pi 4 CPU, Nvidia Xavier CPU / GPU / DLA, Xilinx Artix-7 FPGA) with a $7\times$ spread in execution time and a $60\times$ spread in cold-start time; seeded random client–server topologies over a store-and-forward backbone; and a workload of real production arrivals in which **tasks arrive in peer groups of ten and exchange data with two partners each**, so the cost of a task depends on *where its partners were placed*. That last ingredient is deliberate: three earlier lines of work (§6.1) proved by measurement that in a simulator where every cost is indexed by a *machine*, the optimal placement is pointwise-separable and a graph model has nothing to reason about.

How it is measured. Training labels come from *brute force*: every feasible joint placement of a task batch is simulated to completion and the cheapest one is the label (§3). Evaluation is a *live gate*: a 50,000-event window of the trace replayed on a cell, learned scheduler against Knative, with bars registered before the data exist (§5). The replication unit is an *environment* = (topology, arrival window), sixteen per rung, because the simulator is deterministic and more tasks buy nothing.

Where the answer stands.

- **No learned arm beats Knative at any operating point where Knative is healthy.** At 6 servers / 40 clients on 16 environments the best arm, `gnnedge0`, is $+6.7\%$ *slower* (0/14 checkpoints); at 80 clients $+14.4\%$ (0/16); at 80 servers -1.6% ($p = 0.002$), real and far inside the 5% bar (§6.2, §6.3).
- **The -20.8% headline of 2026-09-18 was an artefact:** an unpaired ratio of medians over four cells, two of them saturated. Paired per cell the arm loses three of the four (§6.2). The record, the reader and the standing answer were corrected on 2026-09-20.
- **The graph arm is the better learner** (it beats its message-passing-off twin -6.8% on 14/14 checkpoints), **but batching relocates waiting rather than removing it:** the 7.1 s a task waits to assemble its peer group comes back as shorter queues and shorter partner-waits, and shortening the window to 2 s moves the time back without changing the total (§6.4). The arm’s genuine gain is 1.05 s of peer exchange per task from co-location; its genuine cost is the concentration that produces.

Two caveats travel with every number: the arrival window every earlier gate used (w_0) is the burstiest of the four the trace offers, and the peer payloads are a synthetic augmentation of a real trace (§2.4).

2 The environment

2.1 Heterogeneous hardware

The simulator (SimPy, discrete-event, single-threaded) models machines with a hardware class, a set of *platforms* (one task at a time each, FIFO queue), a memory budget, local flash storage with capacity-bounded image eviction, and a network map. Three hardware classes are used in the gate cells, assigned round-robin to machines: a Raspberry Pi 4 (four CPU platforms, 8 GB), a Xavier system-on-module (eight CPU platforms, one GPU, one DLA) and a PYNQ board (one FPGA). Each task type carries a *measured* table per platform type (Table 1), not a scaling factor; the platform that runs a task fastest is often the slowest to start.

task type	platform	execution (s)	cold start (s)	memory (GB)	image (GB)
dnn1	rpiCpu	0.0029	0.33	0.066	3.06
dnn1	xavierCpu	0.0010	0.071	0.069	2.99
dnn1	xavierGpu	0.0208	6.17	0.900	2.99
dnn1	pynqFpga	0.0006	0.98	0.000	0.004
dnn2	rpiCpu	0.1684	0.636	0.069	3.06
dnn2	xavierCpu	0.0239	0.096	0.069	2.99
dnn2	xavierGpu	0.0362	6.14	0.900	2.99

Table 1: The measured per-platform tables for the two inference types in the gate workload (data/nofs-ids/task-types.json). A 3 GB image pull at 100 MB/s is 31.3 s and serialises on the machine’s disk.

2.2 Topology

Topologies are generated deterministically from a seed (src/generate_infrastructure.py): client machines and server machines, client–server links drawn independently with probability 0.6 with per-device-pair latency ranges (30–150 ms), a repair step so that every client reaches at least two replica-hosting servers, and a backbone of 12 core routers (1 Gbps, 4 ms per core link, 20 ms access links) through which every logical link is routed. Transmission over the backbone is store-and-forward: each hop is a capacity-one pipe held for bytes/bandwidth, so only *shared* core segments couple two tasks’ transfers (src/placement/network_fabric.py). Figure 1 shows the cell on which the headline was first read.

cc40s9001: 40 clients (bottom) x 6 servers (top), seed 9001, p(link) = 0.6; server degree 19-25; line width = 1/latency

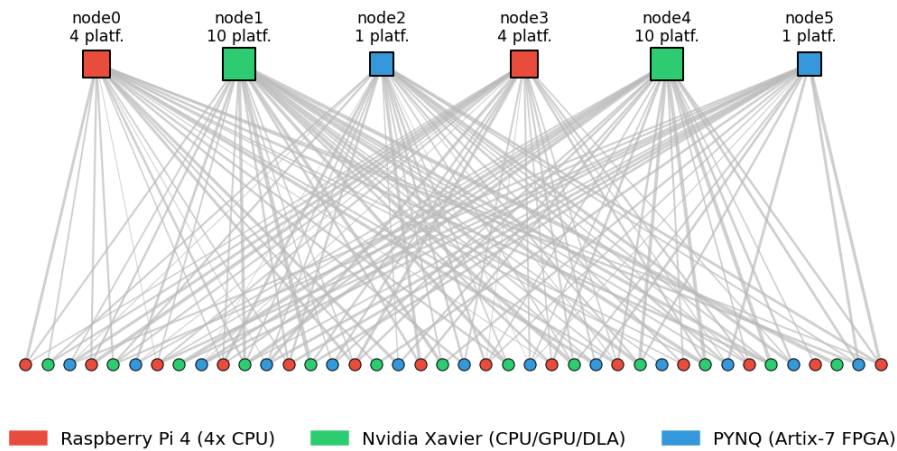


Figure 1: Cell cc40s9001: 40 clients, 6 servers, topology seed 9001. Server squares are sized by platform count; servers host one replica per task type. Adding clients keeps each task’s candidate set inside the training corpus’s support (3–5 candidates per task), adding servers does not (48 at 80 servers).

2.3 How a response time is composed

A task’s response time runs from dispatch to completion. In order, the platform’s processing loop charges: (1) **scheduler wait** — for the learned arms, the time spent assembling the task’s peer group before a decision (§2.4); (2) **FIFO queueing** behind the tasks already on the chosen platform; (3) propagation latency from the source machine; (4) ingress and per-link transmission waits over the backbone; (5) the **image pull** if the machine’s disk does not hold the image, serialised with other pulls on that disk; (6) the platform-specific **cold start** unless the previous task on the platform had the same type; (7) input read plus the **peer exchange** with every partner placed on another machine; (8) execution, a table lookup; (9) output write. Nothing dilates: a co-resident never changes how long a task’s own computation takes; every interaction is a *wait before work starts*. Given infrastructure, workload and placements the loop contains no random draws, so an episode is deterministic — a fact that shapes the evaluation design (§5). Figure 2 shows where the time goes for the three schedulers that matter.

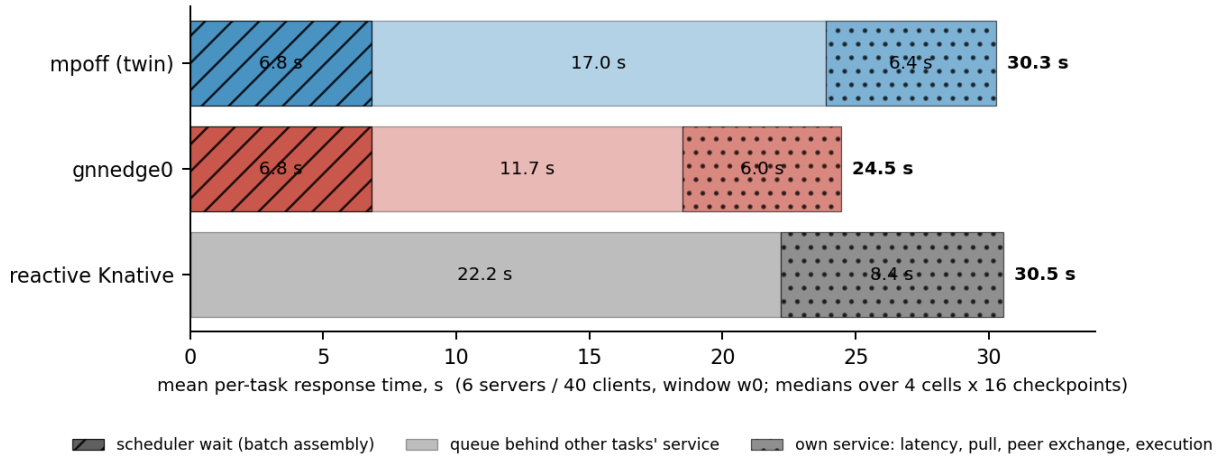


Figure 2: Per-task response time decomposed, 6 servers / 40 clients, window w0. The learned arms pay ≈ 6.8 s of scheduler wait that Knative never pays, and win back more than that in queue; the pointwise twin (mpoff) pays the same wait and does not win it back.

Warmth follows the `node_disk_v2` physics (ADR 0001): a warm sandbox skips the cold start, and only a node-local disk hit skips the pull, so a warm sandbox never masks an evicted image. The earlier `platform_reuse_v1` rule is retained only so that old corpora stay bit-reproducible.

2.4 Why tasks communicate: the peer exchange

The argument. Sections 6.1 and the September 9 report establish, by measurement, that if every contention term is indexed by a machine (queue, disk, memory, ingress pipe), by the task’s own platform (warmth) or by a link on its own route, then co-residents are exchangeable within a task type, every symmetric function of a multiset is a function of its counts, and a pointwise scorer that is told the per-machine counts can express the exact cost. A graph model has no additional fact to work with, and it measurably ties. The one kind of cost that escapes this argument is one indexed by a *pair of task instances* decided jointly: the price of placing i and j apart cannot be written as a function of any machine’s counts.

The mechanism. A workload carries a table of peer pairs (i, j, bytes) . At task i ’s input stage, for every partner j (`src/placement/infrastructure.py, _peer_exchange_time`):

$$c_{ij} = \begin{cases} 0 & \text{if } i \text{ and } j \text{ run on the same machine,} \\ \text{hops}(m_i, m_j) \cdot \frac{\text{bytes}_{ij}}{\min_{\ell \in \text{route}} \text{bw}_{\ell}} + \text{latency}(m_i, m_j) & \text{otherwise,} \end{cases}$$

charged into the task’s service time (so it also inflates the queue of everyone behind it), plus a *rendezvous wait* if j has not been placed yet. Dividing by the receiving machine’s own bandwidth, an earlier form, made

the term separable again — a $100,000\times$ payload scaling probe left additive-argmin regret at exactly 0.000 %; the hop count is what gives distance magnitude. Every legacy corpus is bit-identical with the mechanism off.

The workload, stated honestly. The arrival trace is the project’s own production capture (`workload-150-150.json`: 450,729 single-task events at 150 requests/s over 20 clients, two DNN types), *not* a public dataset. The peer pairs are a *synthetic augmentation* of it:¹ consecutive arrivals are grouped in tens, every task draws two distinct partners inside its group, and each pair’s payload is $200\text{ MB} \times 10^{U(-1,1)}$; the draw is seeded and the provenance (seed, source hash, pair count) is written into the file. The transfer physics is checked; the payload distribution is an assumption, and the report says so wherever a number depends on it. Figure 3 shows the resulting group structure and the fact that decides the scheduler’s cost: a group’s ten members arrive over a median 13.8 s.

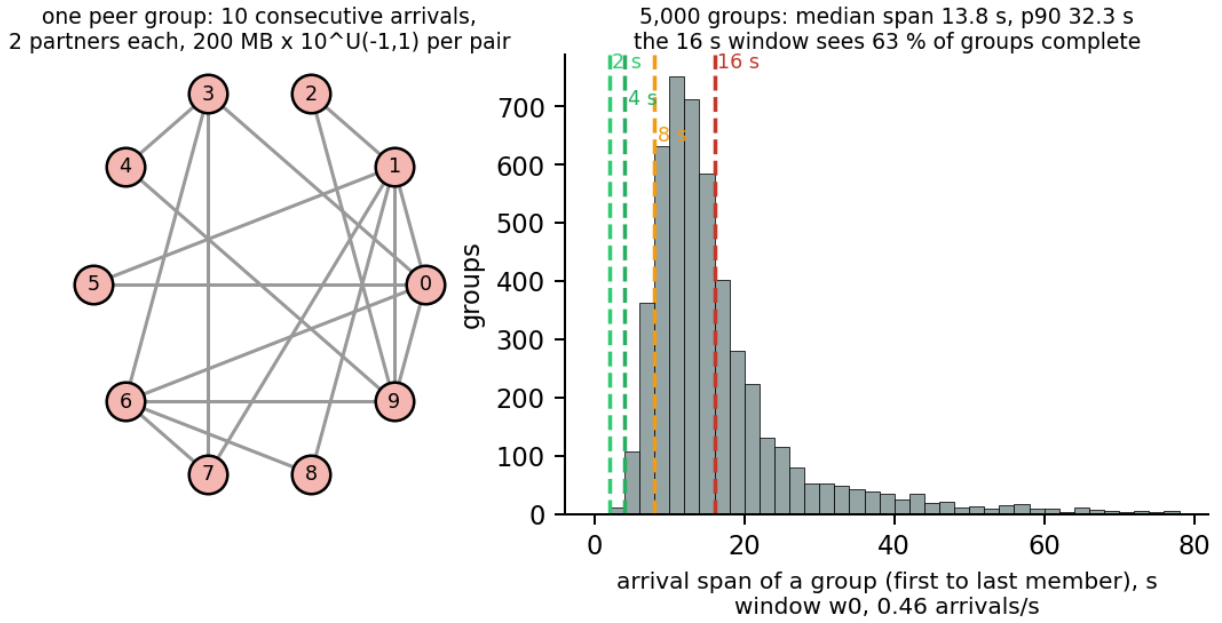


Figure 3: Left: one peer group. Right: the arrival span of the 5,000 groups in the 50,000-event gate window at 0.46 arrivals/s, against the four batch windows studied in §6.4. A learned arm that waits for the whole group pays about half the span per task.

2.5 Is it checked?

- **Determinism.** Seeded topology, replicas, workload draw and training. Re-running a (cell, workload, policy) triple is bit-identical: the reactive run on `cc40s9001` made for tonight’s screen equals the one made two days ago to the last digit of `total_rtt`. A test runs every trainer twice at a fixed seed and requires identical weights.
- **Fail-loud contracts.** Every checkpoint carries a sidecar naming its feature contract, physics, corpus split hash and every architecture flag invisible in the weights; serving refuses a mismatch instead of adopting a default. Every gate script pins, per arm, the flags that separate it from its twin.
- **Registration.** Every lineage commits its bars, reader and tests *before* its data exist, records its expectation, and closes only on a live gate (rule 6 of the record).
- **What the discipline caught.** A batch-timeout precedence bug (a $4,000\times$ window served as 20 ms); a queue feature out of its training range by $300\times$; an unseeded trainer that made every early MLP checkpoint an unreproducible draw; a cluster environment leak that ran three sessions of gates on

¹`scripts_cosim/make_peer_affinity_production_workload.py`, whose usage line builds exactly the gate file.

an undeclared interpreter; and, tonight, 15 topology cells that hang in a known starved-client spin, cancelled and excluded by rule.

3 Brute-force labelling by co-simulation

Supervised training needs, for a cluster state and a batch of tasks, the joint placement that actually minimises the batch’s cost in the simulator. The pipeline (Figure 4; `src/executeecosimulation.py`) produces it by exhaustive enumeration.

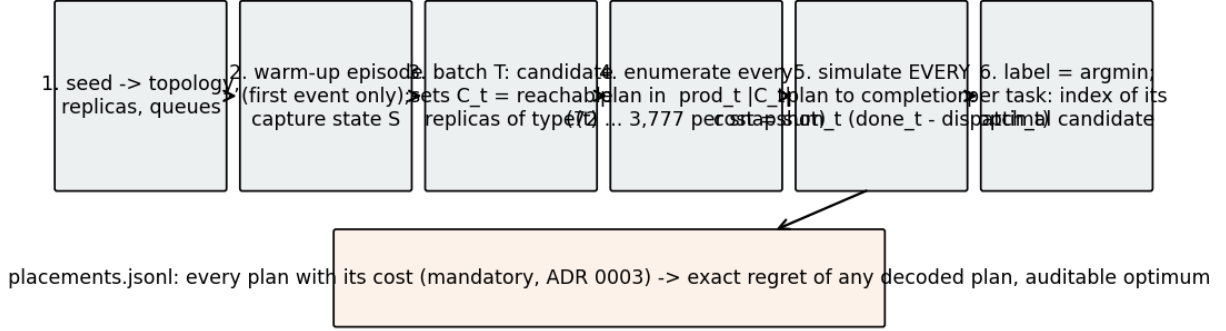


Figure 4: One dataset (snapshot) of the co-simulation corpus.

Let S be the system state after a warm-up episode that runs only the first workload event, and let T be the batch. For each $t \in T$ the candidate set C_t is every (machine, platform) that holds a replica of t ’s type and is reachable from t ’s client. A plan is $\pi \in \prod_t C_t$ (optionally restricted to distinct replicas). Every plan is simulated to completion with the placement forced, and its cost is the episode’s total response time,

$$J(\pi) = \sum_{t \in T} (\text{done}_t(\pi) - \text{dispatch}_t), \quad \pi^* = \arg \min_{\pi} J(\pi),$$

warm-up tasks excluded. The label of task t is the index of π_t^* within C_t . The full sweep $\{(\pi, J(\pi))\}$ is written to `placements.jsonl` and is *mandatory* (ADR 0003): it makes the optimum auditable and the *regret* of any decoded plan, $J(\hat{\pi}) - J(\pi^*)$, an exact lookup. Sweeps hold 72 to several thousand plans per snapshot; datasets whose product exceeds a threshold are skipped with the reason recorded, so a void corpus is never misread. The corpora used here hold 1,670 snapshots (and a 516-snapshot subset for corpus-matched controls), split 1,524 / 96 / 34 by a committed artefact whose hash every checkpoint carries: *a corpus is its split, not its directory*.

Prefix-conditioned labels (`partial_state_v3`). Live decoding commits tasks one at a time, so the corpus is written the same way: for task k the feature row of every candidate carries a 22-dimensional *prefix block* describing what tasks $1..k - 1$ have already committed on that candidate’s machine (type occupancy, committed and remaining demand against capacity, rank of the candidate among its type as a size-free fraction). The earlier 38-dimensional block encoded ranks as a one-hot over at most five machines, which pinned every checkpoint to 6-server cells; the size-free block is what lets a 6-server checkpoint be served on 80 servers at all.

4 The model and its training

Each snapshot is one graph (Figure 5): task nodes (type one-hot, source) and candidate-platform nodes (16 physics features: platform type, queue depth, warmth, remaining work, pull backlog, ...), task–platform edges carrying [execution time, latency, is-warm, energy, communication time], and task–task *peer edges* carrying $\log(1 + \text{bytes}/10^6)$. Two message-passing stages follow the encoders:

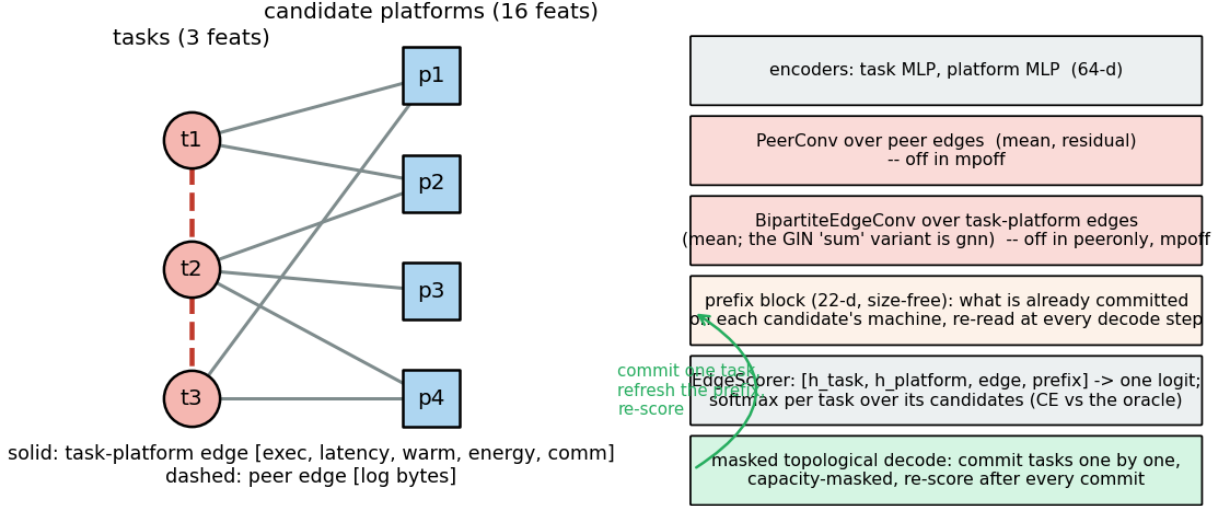


Figure 5: The bipartite graph and the stages of the scoring network. The five arms differ only in which shaded stages are active.

- **PeerConv** over peer edges: $h_i \leftarrow h_i + \text{MLP}([h_i; \text{mean}_{j \sim i} \text{MLP}([h_j; e_{ij}])])$ — residual, mean aggregation.
- **BipartiteEdgeConv** over task–platform edges, same message form, **mean** aggregation over a task’s candidates. The original arm (gnn) used a GIN with *sum* aggregation; a sum over a candidate set that is 3.6 wide at 6 servers and 47.9 at 80 makes the layer a function of deployment size, and that alone cost +13 % at 80 servers and nothing at 6 (bipartite_aggr_v1: SUM-COSTS-ONLY-WHERE-CANDIDATES-ARE-MANY).

An **EdgeScorer** maps $[h_t; h_p; e_{tp}; \text{prefix}_{tp}]$ to one logit per (task, candidate); logits are grouped per task. Decoding is *masked topological*: tasks are committed one by one, candidates masked by machine capacity, the prefix block refreshed and every remaining task re-scored after each commit; there is no unmasked fallback and a failed decode is an error. The same closure computes the prefix at training and at serving time, so the two cannot drift.

arm	PeerConv	bipartite stage	aggregation	role
gnn	on	GIN	sum	the original bipartite arm
gnnedge	on	EdgeConv	mean	the repair, edge attributes on
gnnedge0	on	EdgeConv	mean	the headline arm ; attributes zeroed, matched gnnedge (+0.44 %, 8/16)
peeronly	on	none	—	peer edges alone
mpoff	off	none	—	the pointwise twin: identical weights layout, no message passing

Table 2: The arms. All trained on the same 1,670-snapshot corpus and split, 16 seeds each.

Training (`run_experiment.py` over a YAML per arm; trainer `src/notebooks/train_near_rtt.py`). Per-task cross-entropy over the task’s candidates against the brute-force optimum; AdamW at 2×10^{-3} ; up to 300 epochs with patience 60 and a 100-epoch floor; the checkpoint is chosen by *validation decode regret* (exact, from the sweeps), never the last epoch — selected-vs-last is worth +13–16 % live. Sixteen seeds per arm; every run logs to Weights & Biases; every checkpoint is saved with its contract sidecar. Arms differ by exactly one flag in their YAML, so at a fixed seed gnnedge0 and its twin start from bit-identical weights and differ in one operation.

5 Evaluation design

5.1 The live gate and why 50,000 tasks

A live gate replays a window of the trace on a cell: the policy under test receives arrivals in simulated time, scales replicas, places tasks, and the metric is the mean per-task response time (`averageElapsedTime`). One run is 50,000 events, rescaled to 0.46 arrivals/s (30.2 hours of simulated time, about three minutes of wall clock). Three facts fix the design:

1. **Task count is not a lever.** Because an episode is deterministic, one (environment, checkpoint) run carries *zero* sampling noise. All the variance in a checkpoint’s score is interaction with the environment, and the standard deviation of the checkpoint statistic scales as exactly $1/\sqrt{n_{\text{environments}}}$ (measured: 36.4 $\rightarrow$ 18.1 pp from one cell to four). A longer trace only averages more load regimes into one number; it tightens nothing.
2. **The baseline must be healthy.** Knative’s capacity does not grow with the cluster: 80 servers are unsaturated only at the *same* 0.46 arrivals/s that 6 servers run at (Figure 6). Every earlier “–30 to –47 % at 80 servers” was read against a baseline pushed $13\times$ past a ceiling that never moved. A cell enters a study only if reactive’s queue share (queue / response time) is ≤ 0.80 ; a cell that hangs is *unreadable*, and unknown is not a pass.
3. **The window is not “the workload”.** Every gate before this week used the first 50,000 events of the trace. That window is $1.6\times$ slower natively (the ramp-in) and, once rate-matched, the burstiest of the four the trace offers (Figure 7). Each window therefore gets its own rescale factor so that all four deliver 0.4604 arrivals/s and the window axis is arrival *pattern*, not load.

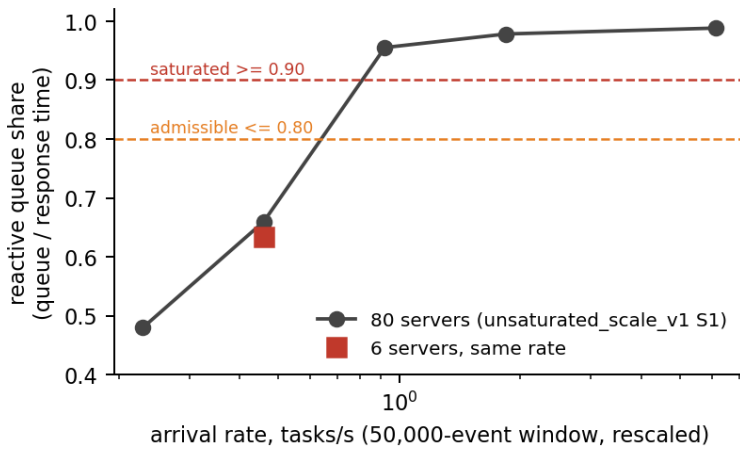


Figure 6: Reactive Knative’s queue share against arrival rate (`unsaturated_scale_v1`). The 80-server cluster is healthy only at the rate 6 servers run at.

5.2 The statistic and the bars

An *environment* is a (topology, window) pair; a study is $4 \times 4 = 16$ of them, fully crossed. The **unit is the checkpoint** (16 per arm). For checkpoint s and environment e let $r_{s,e} = 100 (\text{elapsed}_{s,e} / \text{elapsed}_e^{\text{ref}} - 1)$, the relative difference against the reference *on the same environment*; the checkpoint statistic is $\text{median}_e r_{s,e}$. Arm-vs-arm contrasts pair raw elapsed times on both environment and checkpoint. Every read is a two-sided Wilcoxon signed-rank test of the 16 statistics against zero and fires only if $|\text{median}| \geq 5\%$ and $p < 0.05$; a significant 1.6 % is reported as real and *not separated*. A design-power bar (worst-arm $\text{sd} \leq 10\text{pp}$) is read first, so an under-powered negative is never called a tie. Topologies are selected by the lowest seed ids admissible on every window — a rule no arm can steer. Baselines on every environment: `knative_network` (the reference), `knative_network_ect` (expected completion time), `random_network`. (`knative_network_batch` was measured at 0.002 s of batch wait and is a second reactive variant, not a batching control.)

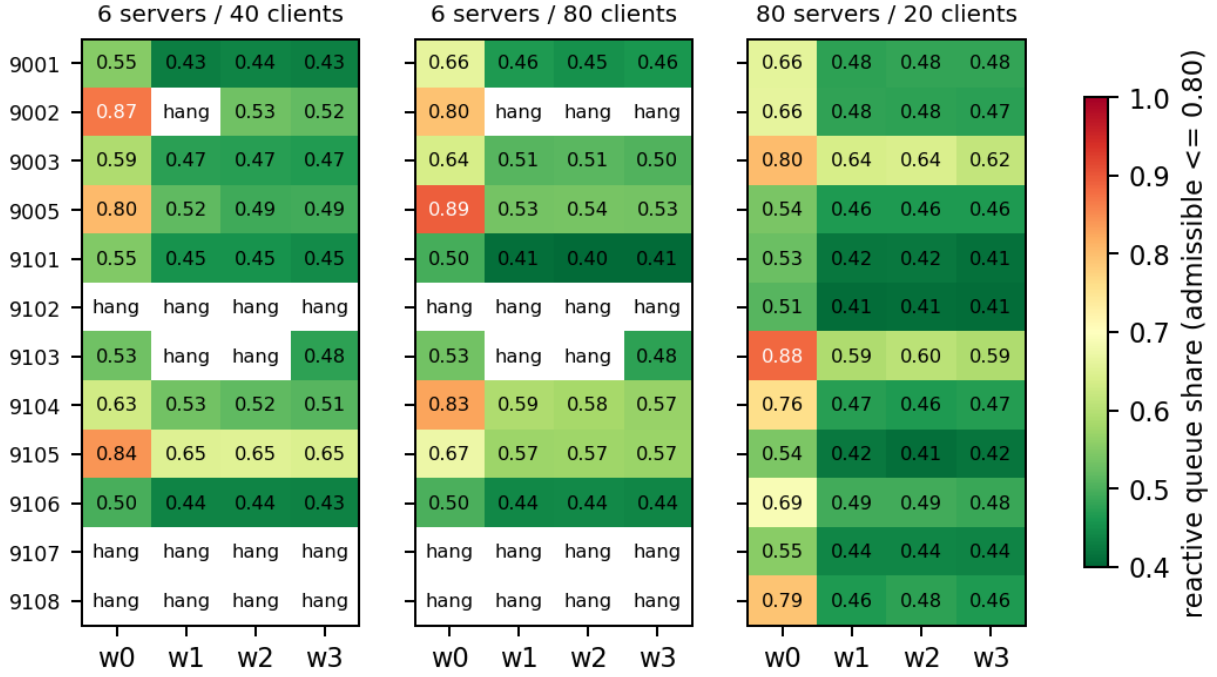


Figure 7: Reactive queue share per candidate topology and arrival window on the three rungs studied (the saturation screens). Every saturated cell is a w0 cell; 15 cells per 6-server rung hang in the starved-client spin (a recorded simulator defect) and are inadmissible. The four lowest-numbered topologies admissible on all four windows form each rung’s study.

6 Results

6.1 What came before, in one section

The programme’s first month (record 2026-08-17 to 09-09, September 9 report) ran three routes to a positive answer and closed all three by measurement. Learned schedulers beat Knative by a wide margin on the original corpus (-44.8% on a held-out fabric), but every advantage of the GNN over a pointwise MLP vanished under a matched control: a pointwise model trained on the same corpus ties the GNN on latency; the reliability edge was 87% a corpus effect and not established once matched ($p = 0.11$, $n = 16$); message passing was a null ingredient on every graph tried ($p = 0.37$, $p = 0.25$). The structural reason (count-shaped costs, §2.4) was corroborated by a one-integer control and by an additive fit with $R^2 = 1.00000$ on plans that place every task on a distinct machine. Two alternatives to the supervised target were measured negative: multi-step horizon labels are deterministic chaos, and closed-loop policy gradient does not improve the supervised checkpoint ($n = 120$, $p = 0.93$). The peer-exchange environment (peer_affinity_v1) followed from that argument, and its first positive was offline: message passing beats its twin by $+5.14$ pp ($p = 0.001$, $13/16$) at 482 snapshots, reproduced at 1,654. Offline edges then had to be taken live, which is where the rest of this section lives.

6.2 The headline at power: 6 servers, 40 and 80 clients

Between 2026-09-16 and 09-18 four lineages (peer_only_v1, bipartite_edge_v1, bipartite_aggr_v1, best_arm_v1) reported, on four cells of window w0 with 16 checkpoints each, that gnnedge0 beats Knative by -20.8% at 40 clients and -13.5% at 80. unsaturated_edge_v1 (registered 2026-09-19, commit 96f0b69; closed 2026-09-20) re-measured the headline on 16 environments per rung with the same checkpoints and bars, and found two things.

The headline was an unpaired statistic. The earlier reader collapsed each checkpoint to the *median of the arm over the four cells* and divided by the *median of Knative over the four cells*. Those two medians

come from different cells; Knative’s time on the four ranged from 18 to 64 s. Read *paired*, cell by cell, the same data give:

40 clients, w0	9001 (share 0.55)	9002 (0.87)	9003 (0.59)	9005 (0.80)	old statistic	paired
gnnedge0	+4.6	+75.8	+9.7	−36.8	−20.3 %	+7.0 %
peeronly	+6.9	+60.4	+11.7	−33.4	−14.9 %	+9.4 %
mpoff (twin)	+13.2	+57.2	+32.0	+13.8	+18.5 %	+22.0 %

Table 3: The four original headline cells, per cell (% vs Knative, medians over 16 checkpoints). The two cells Knative was slow on are the two that fail the saturation bar; the arm wins only on one of them and loses the other catastrophically.

On 16 admissible environments, every learned arm loses to Knative. Figure 8 and Table 4. The design delivered its power (worst-arm sd 1.2 pp at 40 clients, 2.9 at 80, against the 10 pp bar); every arm keeps its sign on all four windows; *gnnedge0* and *gnn* are *unreadable by the letter* at 40 clients because 2 and 4 of their 256 arms hang deterministically in the starved-client spin (the reader refuses a ragged design), and their reads on the complete checkpoints are reported as *disclosed*.

rung	arm	vs Knative	vs random	verdict
40 cl.	gnnedge0	+6.68 %, 0/14, $p = 0.001$	−4.47 %, 14/14	disclosed ($n = 14$)
40 cl.	peeronly	+8.36 %, 0/16, $p = 0.0004$	−1.42 %	REACTIVE-FASTER
40 cl.	mpoff (twin)	+11.53 %, 0/16, $p = 0.0004$	+1.72 %	REACTIVE-FASTER
40 cl.	gnn (sum)	+9.24 %, 0/12, $p = 0.002$	−1.19 %	disclosed ($n = 12$)
80 cl.	gnnedge0	+14.40 %, 0/16, $p = 0.0004$	−0.23 % (tie)	REACTIVE-FASTER
40 cl.	gnnedge0 vs mpoff	−6.82 %, 14/14, $p = 0.001$		GRAPH-FASTER
40 cl.	gnn vs gnnedge0	+2.12 %, $p = 0.01$		SUM-NOT-SEPARATED

Table 4: The registered reads on 16 environments per rung (unit = checkpoint; median over environments of the paired relative %). Knative’s median response time is 17.9 s at 40 clients and 18.3 s at 80; random placement is +4.5 % and +13.9 % behind it.

What survives, and what it means. The graph arm is the better *learner*: it beats its message-passing-off twin on 14 of 14 checkpoints, paired on environment and checkpoint, and the *sum* penalty is absent at 6 servers exactly as *bipartite_aggr_v1* predicted. Where the deficit sits is the subject of §6.4: the arm pays 7.1 s of scheduler wait per task, and the question is whether that wait is a tax on top of a good placement or the same waiting booked elsewhere.

6.3 At scale: 80 servers

unsaturated_scale_v1 found the capacity fact of Figure 6 and, at the one healthy rung, that no arm beats Knative on four cells; *unsaturated_scale_v2* re-read the same five quantities on 16 environments (1,328 arms, 0 failures, worst-arm sd 4.48 pp against the 10 pp design bar). Figure 9 is the finding: **every direction replicated, every magnitude collapsed**, because the four v1 cells were all window w0. *gnnedge0* is −1.62 % ($p = 0.0023$, 13/16), *peeronly* −1.23 % ($p = 0.0013$, 15/16), the GIN-sum arm loses +2.31 %, the matched model-class edge is −2.38 % ($p = 0.0004$) — all real, all inside the 5 % bar. The batch-assembly wait is 7.12–7.13 s for every learned arm and 0.000 s for Knative, ECT and random: 37 % of Knative’s 19.1 s response time, and the arms win back almost exactly that much queue. *random_network* is +8.4 % behind Knative in median with a bimodal tail to +782 %.

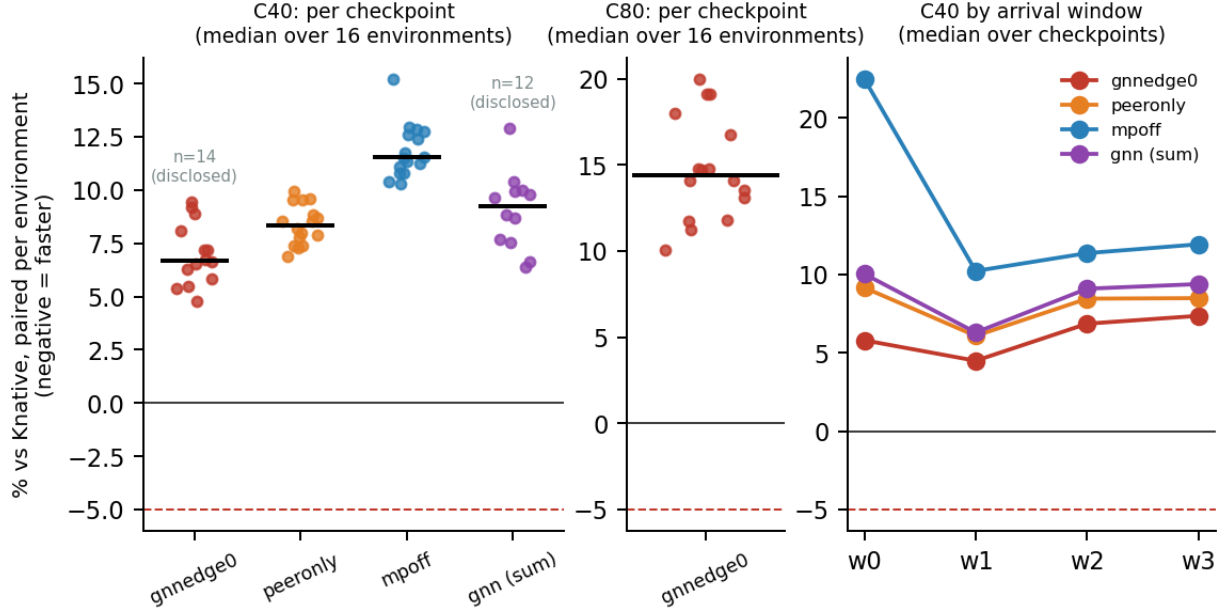


Figure 8: Left, middle: the 16 checkpoint statistics per arm against Knative at 40 and 80 clients (bar = median; dashed = the -5% bar). Right: the 40-client medians by arrival window. Nothing is below zero.

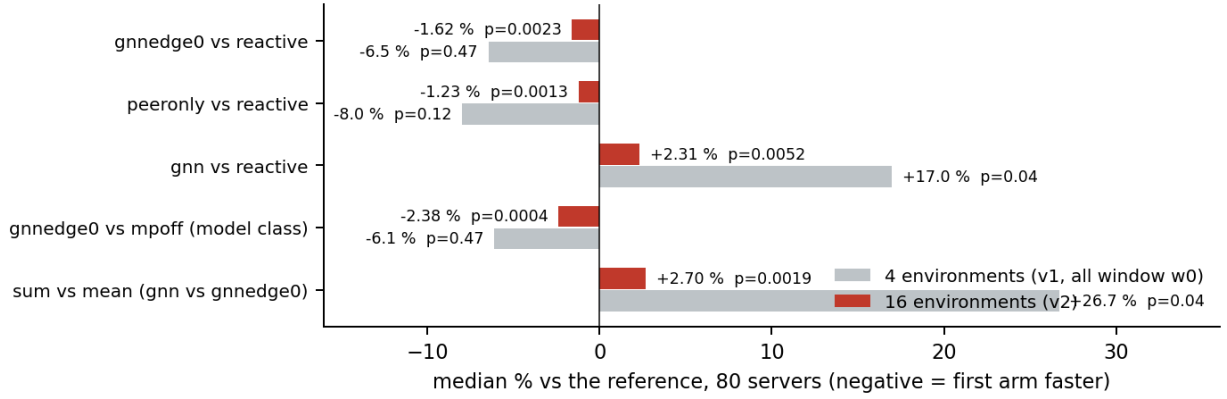


Figure 9: Quote a direction from a small design and a magnitude only from a large one: the five registered quantities at 80 servers, four environments (all w0) against sixteen.

6.4 The batch tax and the window

Every learned arm collects a task's whole peer group before it decides, waiting up to `batch_timeout` = 16 s for members not yet arrived; with group spans of median 13.8 s (Figure 3) that is 6.8 s per task. The window is a policy constant set once and swept once, at the 20-client rung where every learned arm loses to Knative by $\geq 70\%$ whatever the window. `batch_window_edge_v1` (registered 2026-09-19, commit `f6384eb`) screens 2 / 4 / 8 s on the lowest-seed study topology (four windows, 16 checkpoints) and confirms the chosen window on the twelve held-out environments, `gnnedge0` and its twin, with a mechanism read (W6: the wait must actually fall) gating the interpretation.

Result: the window is not the lever. Table 5 and Figure 10. On the screen topology the scheduler wait falls from 7.15 s to 1.66 s as the window shortens from 16 to 2 s, and the median against Knative does not move: +7.70 % at 16 s, +7.87 (4 s), +8.04 (8 s), +8.77 (2 s). The registered rule picks the incumbent (4 and 8 s are ineligible by the letter because one and two of their arms hang in the starved-client spin; the disclosed reads on the complete checkpoints agree), so the lineage closes `WINDOW-NOT-THE-LEVER` on its screen and the held-out phase does not run.

window	wait / task	median vs Knative	w0 / w1 / w2 / w3	queue (w1)	rendezvous (w1)
2 s	1.66 s	+8.77 %	+9.3 / +8.9 / +8.6 / +8.5	8.19 s	3.34 s
4 s	2.65 s	+7.87 % (n=15)	+5.7 / +8.9 / +8.0 / +7.7	7.34	3.20
8 s	4.30 s	+8.04 % (n=14)	+4.0 / +8.7 / +7.9 / +8.1	—	—
16 s	7.15 s	+ 7.70 %	+4.6 / +8.5 / +7.4 / +8.2	4.78	1.2

Table 5: gnnedge0 on the screen topology (cc40s9001, four arrival windows, 16 checkpoints) at four batch windows. What the scheduler releases, the platform queue and the rendezvous wait absorb.

What the decomposition actually says. Execution is ≈ 0.1 s; a task’s time is queueing and peer traffic. Table 6 splits it. The learned arm’s shorter platform queue and shorter rendezvous are *the same waiting relocated*: a task held until its group is complete arrives with its partners already placed (so it does not wait for them at the platform) and finds shorter queues because the other tasks are being held too. The window sweep is the proof — shorten the holding time and those two buckets grow by exactly what the scheduler wait lost. The one bucket that placement itself causes is the **exchange**: co-locating partners saves **1.05 s per task**. The genuine cost is the concentration that co-location produces. The net is +0.4 s per task, and no window can change it, because the window only decides where the waiting is booked. (The earlier reading of this table — “the decisions are good, the waiting is the whole deficit” — is withdrawn in the record.)

40 clients, medians per task (s)	scheduler wait	platform queue	peer exchange	rendezvous	total
Knative	0.00	8.66	5.40	3.69	17.87
random	0.00	10.01	5.05	3.17	18.67
gnnedge0	7.11	5.59	4.35	1.25	18.32
peeronly	7.11	5.99	4.65	1.23	18.80
mpoff (twin)	7.12	6.56	4.77	1.20	19.70

Table 6: The full decomposition on the 16 study environments (execution and I/O, ≈ 0.1 s, omitted). Only the exchange column is a placement effect.

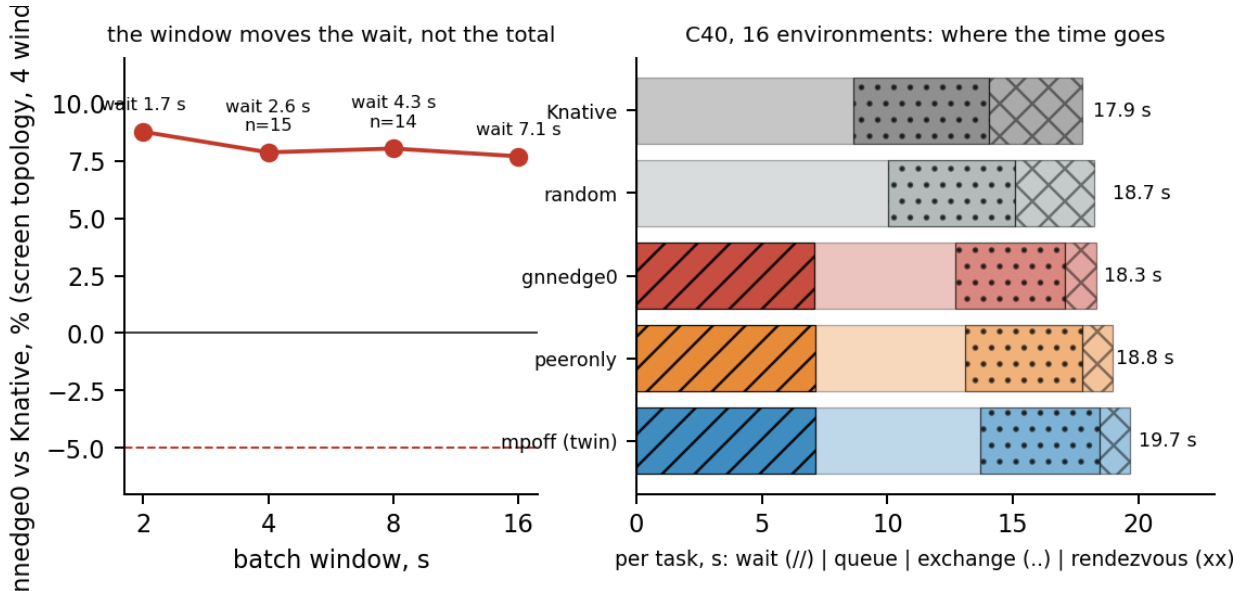


Figure 10: Left: the screen — gnnedge0 vs Knative by batch window. Right: where each arm’s time goes on the 16 study environments; hatched = scheduler wait, dotted = peer exchange, cross-hatched = rendezvous.

7 Limitations and next steps

- **Checkpoints trained at 6 servers.** Every arm was trained on a 6-server corpus; at 80 servers it is served $9.6\times$ out of candidate support. The 80-server null closes “the existing checkpoints at scale”, not “a GNN trained at scale”.
- **Synthetic peer payloads** on a real trace (§2.4). The magnitude of every peer-related effect depends on the 200 MB scale; the direction of the argument does not.
- **The starved-client spin.** 15 of 48 cells per 6-server rung hang; it is a known simulator defect (a postponed task retried every batch when its client’s reachable servers are memory-full), recorded, and excluded by a rule fixed before the data. Fixing it would widen the admissible set, not change a number.
- **w0 dependence.** The headline’s history is a w0 history. Tonight’s runs are the first that can say what it is worth elsewhere.
- **Next.** The window is closed as a lever, and so is removing batching outright (-1731% in an earlier lineage). The one configuration this record has never served is a decoder that places each arrival *immediately*, conditioned on the partners already placed — no group wait, no relocated rendezvous — which the prefix-conditioned decoder supports in form. Whether 1.05 s of exchange per task is worth the concentration it causes is then a question the same apparatus answers in an afternoon. The claim this document can make today is the mechanism, not a margin.

A Reproducibility

- Record: `LINEAGES.md (index) → docs/lineages/<name>.md` (one node per lineage, bars signed before data); `docs/lessons.md`; `docs/hard-stops.md`; `tests/test_record_hygiene.py` (holds the record’s invariants).
- Tonight’s lineages: `docs/lineages/unsaturated_edge_v1.md` (commits 96f0b69, a7d0e72), `docs/lineages/batch_window_edge_v1.md` (f6384eb); readers `scripts_cosim/unsaturated_edge_v1_read.py`, `scripts_cosim/batch_window_edge_v1_read.py` with their tests; cluster scripts under `scripts_cosim/datalab/`.
- This document: `paper/professor_summary/`; every figure regenerates from `make_figures.py` and the summaries under `data/`; the statistics are computed by the registered readers, so the figures show exactly the numbers the record carries.
- Earlier chronology: `paper/herosim_findings_report.tex` (2026-09-09).